HUGO GABRIEL PEREIRA DE ALMEIDA

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I am an Applied Mathematics and AI engineering student actively developing low-latency C++ trading infrastructure and deep learning models for quantitative research

EDUCATION

ESEO — Engineering Diploma, Computer Engineering — *Specialization: Applied Mathematics & Artificial Intelligence*
Dijon / Angers, France | Aug 2022 – Aug 2027 (expected)
Relevant coursework: Applied Mathematics, Probability & Statistics, Numerical Methods, Machine Learning, Deep Learning, Time-Series Analysis, Signal Processing, C/C++ & Python for Scientific Computing

Eiffel High School — French Scientific Baccalauréat, European Section — *Graduated with Honors*
Dijon | Sept 2019 – June 2022
Full math courses in english, python learning

EXPÉRIENCE

SNCF Voyageurs Matériel — *AI Engineer Apprentice, Autonomous Systems*
Le Mans, France | Sep 2025 – Aug 2027

- Engineered a low-latency, multi-sensor obstacle-detection architecture for autonomous trains, designing real-time signal-processing pipelines under strict latency constraints.
- Implemented event-driven data fusion across asynchronous event-based camera, LiDAR, and Radar streams, and prototyped Spiking Neural Networks (SNNs) to optimize temporal pattern-recognition speed on high-frequency sensor data — using Python, C++, and advanced Deep Learning / Computer Vision methods.

International University of Monaco — *Quant Finance / AI Engineer Intern*
Monaco | Jun 2026 – Sept 2026

- Formulated a novel AI-driven framework for large-scale cryptocurrency portfolio optimization, applying Deep Reinforcement Learning and time-continuous architectures (Liquid Neural Networks) to hierarchical asset-allocation problems; results submitted for academic publication.
- Designed an AI-driven equity research platform for the CFA Institute Research Challenge, integrating fundamental valuation and pricing models to automate corporate investment report generation.

NextWise — *Portfolio Trading Simulation — Euronext Trading Game*
Feb 2026 – Apr 2026

- Managed and dynamically reallocated a simulated Euronext equity portfolio over a 3-month competitive trading game, applying systematic reallocation rules to improve risk-adjusted performance.

LEAD CNRS UMR 5022 — *Machine Learning Research Intern*
Dijon, France | Jun 2024

- Designed an end-to-end predictive-modeling pipeline from scratch for a clinical research study on high-dimensional healthcare data, benchmarking four algorithms (Gradient Boosting, SVR, LASSO, Random Forest) under rigorous validation protocols to maximize out-of-sample accuracy while controlling overfitting.
- Co-authored the resulting research as 4th author in Kidney International Reports (peer-reviewed journal).

PROJECTS & RESEARCH

UDS-OB-HFT — Low-Latency Limit Order Book Replay Engine (C++) — *Personal Project - Project link*

- Engineered a C++ market-data replay system reconstructing full Limit Order Book (LOB) dynamics from historical crypto tick data via a custom TCP market-data server.
- Built an accompanying paper-trading client to backtest execution strategies against replayed order-book microstructure, emulating HFT market conditions.

Geo-Commodity-Prop — Commodity Supply-Chain Shock Propagation (GNN) — *Personal Project - Project link*

- Built an event-driven commodity research stack that turns news shocks into graph features (LLM extraction → entity→node map → severity injection on mines/ports/chokepoints), then propagates stress through supply networks with a GNN to produce commodity-level tension scores.
- Linked those scores to a systematic quant layer (signal construction, VectorBT backtests, Sharpe / vol / drawdown diagnostics) to evaluate whether geopolitically induced supply disruptions translate into tradable alpha across metals, energy, and agriculture.

CEX-Crypto Trading Stack — Crypto Derivatives Pricing & Systematic Trading Framework — *Personal Project — Private Project*

- **Engineering pricing models** for structured products (DCI) and dual-curve bootstrapping estimators to resolve funding basis, while predicting microstructure execution costs via boosted quantile regressions.
- **Co-architecting** an end-to-end quantitative derivatives pricing and systematic trading monorepo in a 3-person team, bridging a custom C++ LOB replay engine with a Python research environment via pybind11.

Quant-Strat — Systematic Quantitative Strategy Research Repository — *Personal Project - Project link*

- Built a modular research pipeline (hypothesis formulation, feature engineering, model implementation, backtesting, performance analysis) to design and evaluate quantitative trading strategies in Python.
- Implemented an XGBoost-based return-prediction strategy using lagged returns, rolling statistics, and technical indicators, benchmarked via Sharpe ratio, volatility, and drawdown.

PUBLICATION

"Deep Learning Morphometric Analysis on Protocol Biopsies Predicts Future Graft Function" — *Publication link*
Kidney International Reports — peer-reviewed | 4th author

TECHNICAL SKILLS & LANGUAGES

Programming: C / C++ (low-latency systems), Python (NumPy, pandas, scikit-learn, PyTorch)
Quant / ML Methods: Machine Learning, Deep Learning, Deep Reinforcement Learning, Recurrent & Liquid Neural Networks, Spiking Neural Networks, Time-Series Analysis, Signal Processing, Computer Vision
Mathematics / Modeling: Applied Mathematics, Probability & Statistics, Portfolio Optimization, Equity Valuation & Pricing Models, Algorithm Benchmarking, Overfitting Control & Validation Protocols
Tools: Git/GitHub, TCP/IP Networking, Market Microstructure & Limit Order Books
Certifications: J.P. Morgan Quantitative Research Job Simulation (Forage) · Getting Started with Deep Learning · Cambridge Certificate in Advanced English (CAE) · TOEIC (925)
Languages: Portuguese (Native) · French (Bilingual) · English (Full Professional) · Spanish (Limited Working)